MILITARY SPECIFICATION

TEST FACILITIES KIT, MK-994( )/AR
(UNITS OF)

1. SCOPE

1.1 This specification covers one type of Maintenance Kit which provides facilities for troubleshooting, bench testing of communications, communication system control, and direction finding equipments and is designated, Test Facilities Kit, MK-994( )/AR. (See 6.1 & 6.3)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585	Box, wood, wirebound style 2 or 3
PPP-B-601	Box, wood, cleated plywood
PPP-B-621	Boxes, wood; Nailed and Lock-corner
PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triplewall
PPP-C-843	Cellulosic Cushioning Material
PPP-F-320	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes
PPP-S-760	Strapping; Non-Metallic
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, (For Carton Sealing)
PPP-T-97	Tape; Pressure-Sensitive, Filament Reninforced
QQ-S-781	Strapping, Flat Steel

MILITARY

MIL-P-116	Preservation, Methods of
MIL-P-3938	Pallet, Material Handling Hardwood Stringer Construction, 4 way (partial)
MIL-P-15011	Pallet, Material Handling, Wood, Post Construction, 4 way entry

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-S-19500	Semiconductor Device, (General Specification for)

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-108	Definitions of and Basic Requirement for Enclosure for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit loads; (40 in x 48 in 4 ways partial and 4 way pallets)
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods

DRAWINGS

ELECTRONIC COMMAND

SC-DL-595579	Test Facilities Kit, MK-994( )/AR
SC-DL-595580	Panel Assembly, Test Facilities Kit SB-3202( )/AR
SC-DL-595581	Cover, Test Facilities Kit, CW-986( )/AR

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Description. - The Test Facilities Kit, MK-994( )/AR is a light weight test set contained in a spray-tight combination case and provides the facilities and connections required for operational and maintenance tests for Radio Sets AN/ARC-114( ), AN/ARC-115( ), and AN/ARC-116( ), Control Communication System C-6533( )/ARC and Direction Finder Set AN/ARN-89( ). Connector and cable assemblies, etc., are not exposed when the cover is in place. The test set contains provisions for a Control, Communication System C-6533( )/ARC and an Indicator, Heading-Radio Bearing ID-1351( )/A, these units are part of the test set. The test set operates from 115 VAC, 400 Hz and +27.5 volts dc power sources.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.2 Preproduction samples. - The contractor shall furnish three (3) preproduction samples for approval, if required by the invitation for bids and contract. (See 4.3 and 6.2)

3.3 Construction. - The equipment shall be constructed in accordance with the following data lists and drawings thereon:

SC-DL-595579	Test Facilities Kit, MK-994( )/AR
SC-DL-595580	Panel Assembly, Test Facilities Kit SB-3202( )/AR
SC-DL-595581	Cover, Test Facilities Kit, CW-986( )/AR

3.4 Cleaning. -

3.4.1 Parts. - After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.4.2 Units. - After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.5 Controls. - All movable controls shall operate properly without binding or other undue restriction. Controls shall not be assembled in a misaligned condition.

3.6 Finish, protective. - The equipment shall be given protective finish in accordance with Specification MIL-F-14072 and the equipment drawings. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch-up after mounting. The final paint film on Type 1 surfaces shall be semi-gloss, light gray (Film E) conforming to Specification MIL-F-14072. (See 4.4)

3.7 Securing of parts. -

3.7.1 General. - Brackets, lugs, inserts, bolts, and other mounting arrangements shall retain items securely when subjected to specified service conditions.

3.7.2 Securing of parts by threaded fasteners. - Nylok or other screws with plastic devices and Loctite or similar sealants shall not be subjected to temperature in excess of 250°F (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

3.8 Marking. -

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.8.1 General. - Marking shall conform to Specification MIL-M-13231. Front panel marking shall be Group I as described in that Specification. (See 4.4)

3.8.2 Visibility. - Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.8.3 Serial numbers. - The following items shall be serial numbered:

Test Facilities Kit	MK-994( )/AR
Panel Assembly, Test Facilities Kit	SB-3202( )/AR
Cover, Test Facilities Kit	CW-986( )/AR

3.9 Soldering. -

3.9.1 Solder. - Solder shall conform to the equipment drawings.

3.9.2 Acid or acid salts. - No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.9.3 Process. - There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystalized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the unit. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.10 Wiring and cabling. - Wiring and cabling shall be neat and sturdy.

3.10.1 Cabling. - Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.10.2 Slack. - Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

a. To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

b. To enable parts to be removed and replaced during servicing without disconnecting other parts.

c. To provide for at least two replacements of the parts to which the wire or cable is connected, except where uninsulated wires are used.

3.10.3 Protection. - Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges.

3.10.4 Support. - Wire and cable shall be properly supported and secured to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (i) during and after subjection of the equipment to specified service conditions or (ii) after service or repair of the equipment in a normal manner.

3.10.5 Clearance. - Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.10.6 Splicing. - Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.10.7 Connections. -

3.19.7.1 General. - Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.11 Semiconductors (transistors and diodes). - These electron devices (semiconductors) shall be in accordance with MIL-S-19500 and the equipment drawings. (See 4.4)

3.12 Tropicalization of materials. - Materials shall be treated as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.12.1 Treating materials. - Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treated material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated materials will be held to the practical minimum.

3.12.2 Toxicity. - Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.12.3 Flexibility. - Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.12.4 Statement of treatment. - The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2)

3.13 Service conditions (See 4.8). - The MK-994( )/AR shall meet the following service conditions -- where a test is referenced, meeting the test shall be considered as compliance with the requirement:

3.13.1 Temperature (See 4.8.1). -

3.13.1.1 Operating. - Ambient temperature in the range of +100°F to 0°F. Exposure at the high temperature extreme not to exceed 4 hours, and at the low temperature extreme not to exceed 72 hours, at any one time.

3.13.1.2 Non-operating. - Exposure in the range of +160°F to -65°F; exposure at the high temperature extreme is not to exceed 4 hours, and at the low temperature extreme not to exceed 24 hours, at any one time.

3.13.2 Humidity (See 4.8.2). - The equipment shall meet the requirements of Procedure 1, Method 507, of MIL-STD-810.

3.13.3 Altitude (See 4.8.3). -

3.13.3.1 Operating. - Up to 10,000 feet above sea level.

3.13.3.2 Non-operating. - Up to 50,000 feet above sea level.

3.13.4 Spray-tight protection (See 4.8.4). - The equipment shall meet the Table I conditions of MIL-STD-108 for "spray-tight" enclosure. There shall be no evidence of moisture inside the equipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.13.5 Fungus (See 4.8.5). - The unit(s) shall show no evidence of viable fungus or corrosion after being subjected to 28 days of fungus test.

3.13.6 Vibration (See 4.8.6). - The equipment shall meet the requirements of Test Method 514-1 and Figure 514-6 (Curve AB) in MIL-STD-810.

3.13.7 Shock (See 4.8.7). - The equipment shall not suffer damage or fail to operate after subjection to the requirements of Table I, as follows:

TABLE I

Paragraph Heading	Paragraph No.	Performance After Test
Shock; bench-handling	4.8.7.1	Specified performance (Note A) No physical damage.
Shock; drop	4.8.7.2	Operable (Note A). Any physical damage shall be minor only.

NOTE A: The equipment shall meet the requirements listed in 4.8.

3.14 Electrical requirements (See 4.9). -

3.14.1 Continuity, relay (See 4.9.1). - The resistance between the indicated points shall be not more than two (2) ohms when dc circuit breaker CB2 is on and the switches are in position shown:

CONTINUITY		SWITCH and POSITION	
FROM	TO		
J1-Z	J1-E	Radio Antenna Function (S1) Headsets (S6) Radio Test (S2)	XCVR TRANSMIT OFF
J1-Z	J1-E	Radio Antenna Function (S1) Radio Test (S2)	XCVR 4
J1-f J2-f	J1-j J2-j	Radio Test (S2)	5

3.14.2 Continuity, connectors (See 4.9.2). - The resistance between the indicated points shall be as shown when the switches are set to the position indicated:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

CONNECTORS		RESISTANCE		SWITCH and POSITION	
From	To	Min	Max		
J3-C	J19-A	4545	5355	COMM CONT	1
J3-d	J3-Z	135.0	165.0	Test (S3)	
J3-e	J3-Z	135.0	165.0		
P1-EE	J3-Z	135.0	165.0		
P1-VV	J3-Z	589.0	651.0		
J3-i	J3-Z	135.0	165.0		
J3-g	J3-Z	135.0	165.0		
J3-h	J3-Z	589	651		
P1-WW	J3-Z	135.0	165.0		
J3-r	J3-Z	135.0	165.0		
J3-a	J3-Z	135.0	165.0		
J3-f	J3-Z	135.0	165.0		
J16-C	J3-C	4.64	5.66		
J16-B	J16-D	7.40	9.00		
J9-CT	Chassis	45.0	55.0	Circuit Breaker (CBI)	ON
J27-A	J27-B		20	ADF Test (S4)	6
J21-A	J21-B		20	Radio Test (S2)	10
J1-M	J1-P	990.0	1010.0	Radio Test (S2)	11
J1-W	J1-P	990.0	1010.0	Radio Test (S2)	10
J18-A	J1-M		2.0	Radio Test (S2)	11
J18-A	J1-W		2.0		

3.14.3 Output voltage, connectors (See 4.9.3). - The output voltages shall be 6.8 ± 0.5 volts dc between pins A and B of J21, and 27.5 ± 1.0 volts dc between pins D and E of connector J1. The output voltage between pins A and B of connector J21 when switch S4 is in position 6 shall be 26.0 ± 2.0 volts ac, 400 Hz.

3.14.4 Lamp, control signal (See 4.9.4). - When each indicated circuit is grounded and each switch is placed to Position 2, the control signal lamp shall illuminate.

CIRCUIT GROUND	at	CONNECTOR	
Retransmit Control (output)	J1	Radio Test (S2)	
Retransmit Control (input)	J2	Radio Test (S2)	
XCVR NO. 2 Transmit Control	J3	COMM CONT TEST (S3)	
XCVR NO. 4 Transmit Control	J3	COMM CONT TEST (S3)	
XCVR NO. 3 Transmit Control	J3	COMM CONT TEST (S3)	
XCVR NO. 1 Transmit Control	J3	COMM CONT TEST (S3)	
XCVR NO. 5 Transmit Control	J3	COMM CONT TEST (S3)	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.14.5 Headsets and lamp, control signal (See 4.9.5). - When the Headsets switches are placed in TRANSMIT position and the TRANSMIT control circuits are jumpered as specified, the control signal lamp shall illuminate when the COMM CONT test switch is placed to the positions specified.

3.14.6 Control, C-6533( )/ARC (Rec) (See 4.9.6). - With a 1000 Hz input signal applied to the interphone line of COMM CONT No. 1 and the output of the headset amplifier measured at HEADSETS Connector J15, terminated with an 8 ohm load set at 1.25 volts, the input signal shall be between 1.00 and 2.75 volts. An input of 300 Hz and 6000 Hz shall not cause a variation of more than +1.0 or -2.0 db from the input value.

3.14.7 Control, C-6533( )/ARC (Trans) (See 4.9.7). - With an input signal level of 0.60 volts at 1000 Hz, from a 5k to 5 ohm impedance matching network, connected to the microphone amplifier through the HEADSETS 1, connector J15, the COMM CONT No. 1 shall produce an output of 2.45 to 3.1 volts ac on the interphone line. An input of 300 Hz and 6000 Hz shall not cause a deviation of more than +1.0 or -2.0 db from the output value.

3.14.8 Radio antenna (S1) XCVR, atten. (See 4.9.8). - With signal inputs of 30, 50 and 400 MHz, 50% modulation at 1000 Hz, and an attenuation of at least 10 db, applied at the Signal Input (J12) and Radio Antenna Function Switch placed to XCVR position, the output level at T/R Antenna (J11) shall be attenuated 22.0 ± 1.0 db.

3.14.9 Switch (S1) radio antenna function (See 4.9.9). - With input frequencies, 50% modulation at 1000 Hz and attenuated at least 10 db, applied at the Signal Input (J12), switch S1 set to the positions indicated, T/R Antenna (J11) terminated into 50 ohms, and Signal Outputs J13 and J14 terminated as indicated, the output attenuation shall be as follows:

Freq. MHz to J12	Termination 50 ohms	Switch S1 Positions	Output Atten. J13	J14
30, 80	J13	HOMING LEFT		
30, 80	J14	HOMING LEFT	3.0 ± 1.0 db below J14	Record
30, 80	J14	HOMING RIGHT	Record	
30, 80	J13	HOMING RIGHT		3.0 ± 1.0 db below J13 *
30, 80	J13	HOMING BALANCE		
30, 80	J14	HOMING BALANCE	*	

* Outputs shall be within ± 0.5 db.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

3.14.10 ADF Antenna, atten. (See 4.9.10). - When input signals of 100 kHz and 3.0 MHz, 50% modulation at 1000 Hz and an attenuation of at least 10 db, applied at Signal Input (J23) positioning the ADF antenna function test switch (S5) to COMP, LOOP and X shall provide an output attenuation of - 9.5 ± 1.0 db at the LOOP ANTENNA X output jack (J24) when the SENSE ANTENNA (J26) and LOOP ANTENNA Y (J25) output jacks are terminated into 50 ohms. Connecting the load across the LOOP ANTENNA X (J24) and SENSE ANTENNA (J26) output jacks shall provide an output attenuation of - 9.5 ± 1.0 db at the LOOP ANTENNA Y (J25) output jack. Connecting the load across the LOOP ANTENNA X (J24) and & (J25) output jacks shall provide an attenuation of - 12.5 ± 1.0 db at the SENSE ANTENNA (J26) output jack with ADF antenna function test switch in the COMP and ANT position.

3.14.11 T/R Antenna, atten. (See 4.9.11). - With signal inputs from 30 to 400 MHz applied to T/R Antenna (J11), and with Radio Test switch in Position 4, the losses between J11 and dummy Load (AT1) shall not exceed 1.0 db.

3.15 Bounce preconditioning (See 4.6). - The equipment shall be capable of meeting the requirements specified herein, without subsequent processing, after subjection to the bounce preconditioning.

3.16 Operational inspection (See 4.12). - The test set shall be capable of meeting the specified test after applicable processes, inspection and test have been completed to determine that each test set on order is operable prior to packaging.

3.17 Interchangeability (See 4.11). - Like units, assemblies, sub-assemblies, modules and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand-picked for fit or performance; however, matched pairs of sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.18 Technical literature and spare parts. - Technical literature and running spares and spare parts shall be furnished if specified in the contract. Spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2)

3.19 Workmanship (See 4.10). - The equipment shall be manufactured and assembled in accordance with the equipment drawings and requirement nine (9) of MIL-STD-454 and the following paragraphs herein:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

Cleaning	3.4
Controls	3.5
Securing of Parts	3.7
Soldering	3.9
Wiring and Cabling	3.10
Tropicalization of Materials	3.12

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commerical laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection. - Inspection shall be classified as follows:

a. Preproduction inspection (does not include preparation for delivery.) (See 4.3)

b. Inspection covered by subsidiary documents. (See 4.4)

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.13 and 4.14)

4.3 Preproduction inspection. - This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4 and the inspection specified for Group A, Group B and Group C, (See Tables II and III and IV respectively). The preproduction inspection will normally be performed in this order: (1) altitude, (2) temperature, (3) vibration, (4) humidity, (5) shock, (6) shock, bench handling and (7) shock, drop. Other preproduction inspection may precede, follow, or be interspersed between the foregoing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.4 Inspection covered by subsidiary documents. - The following shall be inspected under the applicable subsidiary documents as part of the inspection of the units before preparation for delivery:

Item	Where Required
Finish	3.6
Marking	3.8
Semiconductors	3.11

4.5 Quality conformance inspection of equipment before preparation for delivery. - The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.6. This does not relieve the contractor of his responsibility for performance of any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government, at its discretion, may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.7) Test equipment for Government verification inspection shall be made available by the contractor. Each unit shall be bounce preconditioned after final assembly and shall not be considered as a portion of acceptance inspection. (See 3.15)

4.5.1 Group A inspection. - Each unit of each lot of equipment shall be inspected for conformance to all the inspection and test requirements of Table II. Each lot will be subject to verification utilizing the procedures of MIL-STD-105 using the general inspection levels and the AQL's indicated in Table II. Group A inspection shall be performed in any order which is satisfactory to the Government, except that the operational inspection 4.12 shall be last (See 6.6).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

TABLE II - Group A Inspection

Inspection	Reqt Para	Insp Para	AQL Major	AQL Minor
Visual and Mechanical:	3.19	4.10		
Circuit Card Assembly			0.65%	2.5%
Cable Assemblies (assembled)			1.0%	4.0%
Cover, CW-986( )/AR			1.0%	4.0%
Extractors, MX-8128( )/AR and MX-8129( )/AR			1.0%	4.0%
Panel Assembly, SB-3202( )/AR			1.0%	4.0%
Facilities Kit, MK-994( )/AR (assembled)			1.0%	4.0%
Electrical:				
Continuity, Relay	3.14.1	4.9.1		
Output Voltage				
Connectors	3.14.3	4.9.3		
Lamps, Control Signal	3.14.4	4.9.4	*1.0%	
Radio Antenna (S1)			for the	
XCVR, Atten	3.14.8	4.9.8	combined	
Switch (S1) Radio			Group	
Antenna Function	3.14.9	4.9.9		
Operational Inspection	3.16	4.12	0.1%	

* All electrical defects shall be considered major defects.

4.5.2 Group B inspection. - This inspection, including sampling shall conform to Table III and to the procedures for small-sample inspection of Standard MIL-STD-105 using the special inspection levels. The AQL shall be 6.5 percent defective and the inspection level shall be S4 for normal, tightened and reduced inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have met the Group A inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

TABLE III - Group B Inspection

Inspection	Reqt Para	Insp Para	AQL
Continuity, Connectors	3.14.2	4.9.2	
Headsets and Lamp, Control Signal	3.14.5	4.9.5	6.5% for the combined Group
Control C-6533( )/ARC (Rec)	3.14.6	4.9.6	
Control C-6533( )/ARC (Trans)	3.14.7	4.9.7	
ADF Antenna, Atten.	3.14.10	4.9.10	
T/R Antenna, Atten.	3.14.11	4.9.11	
Interchangeability	3.17	4.11	6.5%

4.5.3 Group C inspection. - This inspection shall be as listed in Table IV, and shall normally be performed on sample units that have been subjected to and met Group A and Group B inspection. No defects shall be allowed. Sample unit(s) shall be selected in accordance with 4.5.4.

TABLE IV - Group C Inspection

Inspection	Reqt Para	Insp Para
Subgroup I:		
Altitude	3.13.3	4.8.3
Temperature	3.13.1	4.8.1
Vibration	3.13.6	4.8.6
Humidity	3.13.2	4.8.2
Subgroup II:		
Spray Tight Protection	3.13.4	4.8.4
Shock, Bench Handling	3.13.7	4.8.7.1
Shock, Drop	3.13.7	4.8.7.2
Subgroup III:		
Fungus	3.13.5	4.8.5

4.5.4 Sampling for inspection of equipment. - Units selected for each Group C inspection shall be selected without regard to their quality in accordance with the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.5.4.1 Subgroup I. - For this subgroup, two units from the first 10 production units shall be selected for the first Group C inspection. For subsequent Group C inspections, one unit from each successive 25 or fractions thereof shall be selected.

4.5.4.2 Subgroup II. - For this subgroup, two units from the first 10 production units shall be selected for the first Group C inspection. For subsequent Group C inspection, one unit from each successive 50 shall be selected or fraction thereof shall be selected.

4.5.4.3 Subgroup III. - For this subgroup, only one (1) unit(s) from the first 10 production units shall be selected for the first Group C inspection. For subsequent inspection one (1) unit(s) shall be selected when one half of the units on order have been accepted by the Government's QA Representative.

4.5.5 Noncompliance. - The contractor shall immediately report in writing each Group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (i) the process and materials, as applicable, and (ii) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected to Group C inspection (all inspections, or the inspections which the sample failed, at the option of the Government) and Group A and B inspection may be reinstituted; however, final acceptance and shipment will be withheld until the Group C reinspection results have shown that the corrective action was effective. (See 6.4)

4.5.6 Reinspection of conforming Group C sample units. - Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be delivered on the contract or order, provided all damage is repaired, cleaned of all foreign matter, and the sample units are re-subjected to and pass Group A inspection.

4.6 Bounce preconditioning. - The Facilities Kit shall be placed in its normal operating position on a L.A.B. ARV-30-200 vibration table or suitable equivalent. The unit shall be secured to the vibration table and subjected to a two (2) minutes vibration. The vibration force shall be approximately 1g at 25 Hz. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to acceptance inspection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.7 Standard test conditions. - Unless otherwise specified, all inspection shall be performed at normal room temperature, pressure, and humidity under the following conditions:

Input Voltages	27.5 ± 0.5 volts dc 115 ± 5.0 volts ac rms 400 Hz, Single phase
RF Impedance	52.0 ± 2.5 ohms resistive
Standard Test Signal	1000 Hz, modulated 50%

4.8 Service conditions. - The service condition tests shall be conducted as specified. When electrical tests are required, the equipment shall meet the requirements of 3.14.1, 3.14.3 and 3.14.6. Upon completion of each service condition test, the equipment shall be subjected to and pass a visual and mechanical inspection. No defects will be allowed.

4.8.1 Temperature (See 3.13.1). -

4.8.1.1 Operating. - The equipment shall be subjected to the temperature limits in accordance with Methods 501 and 502 of MIL-STD-810. Tests shall be conducted at ambient before and after the temperature tests, and at extreme temperatures when the temperature limits have been reached. The requirements of 3.13.1.1 shall be met.

4.8.1.2 Non-operating. - The equipment shall be subjected to the temperature limits in accordance with Methods 501 and 502 of MIL-STD-801. Electrical tests shall be conducted at ambient before and after the temperature tests. The requirements of 3.13.1.2 shall be met.

4.8.2 Humidity. - The equipment shall be subjected to humidity test in accordance with Procedure 1, Method 507 of MIL-STD-810. Electrical tests shall be conducted within one (1) hour after the completion of the humidity test. The requirements of 3.13.2 shall be met.

4.8.3 Altitude. - The equipment shall be subjected to the altitude tests in accordance with Procedure 1, Method 500, of MIL-STD-810. Electrical tests shall be conducted before and after the altitude test at ambient pressure and at 10,000 feet. The requirements of 3.13.3 shall be met.

4.8.4 Spray-tight protection. - The equipment shall have CW-986( )/AR in place and be subjected to the tests specified in Table 1 of Specification MIL-STD-108, except the equipment shall be sprayed as normal rainfall, for compliance with paragraph 3.13.4.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.8.5 Fungus. - The unit(s) including accessories shall be subjected to 28 days of the fungus test specified in Procedure 1, Method 508 of MIL-STD-810. For gasket or other non-hermetically sealed equipment, the parts and materials of the chassis shall be sprayed with the spore suspension. The interior of the case shall also be sprayed. The unit(s) shall be subjected to the test in this open condition and as specified for compliance with 3.13.5; the unit(s) shall not be required to operate.

4.8.6 Vibration. - The equipment shall be subjected to the vibration test in accordance with Method 514-1 of MIL-STD-810, except the equipment shall not be operated during vibration. Electrical tests shall be conducted before and after the vibration test. With the equipment mounted in the test fixture on the vibration table, determine resonant modes of the maintenance kit by varying the frequency of applied vibration slowly through the specified range at an amplitude not exceeding that for which the equipment is to be tested. Individual resonance surveys shall be conducted with vibration applied along each of the three orthogonal axes of the equipment. Electrical tests shall be conducted after the completion of the vibration tests. The requirements of 3.13.6 shall be met.

4.8.7 Shock. -

4.8.7.1 Shock, bench-handling. - The chassis-with-front-panel assembly shall be removed from its combination case, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing:

a. Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

b. Repeat "a" with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing. The requirement of 3.13.7 shall be met.

4.8.7.2 Shock, drop. - The equipment in its combination case shall be subjected to 18 impact shocks of 15 g. consisting of 3 shocks in opposite directions along each of 3 mutually perpindicular axes, each shock impulse having a time duration of 11 $\pm$ 1 milliseconds. Except for abrasion, paint chipping, and small local deformations, no damage shall occur. Upon completion of the test, without adjustment or repair, the equipment shall be tested for compliance with 3.13.7.

4.9 Electrical requirements. -

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.9.1 Continuity, relay. - Apply 28 volts dc to J28 and place CB2 to ON position and connect P1 to P2, and measure the continuity with the switches placed in the positions specified to determine compliance with 3.14.1.

4.9.2 Continuity, connectors. - Set each switch to the positions specified as the resistance is measured between the points indicated to determine compliance with 3.14.2.

4.9.3 Output voltage, connectors. - Apply +27.5 volts dc to J28 and 115 vac, 400 Hz, to J27 of the equipment and place circuit breakers to ON. Measure the dc voltage at the RADIO SET NO. 1 connector J1-D (Pos) and J1-E (Neg) and the ac voltage at ADF RCVR TEST connector J21-A and B. These measurements shall be made to determine compliance with 3.14.3.

4.9.4 Lamp, control signal. - Apply 27.5 volts dc to J28, set the switches to each position specified and ground the circuits indicated. As each circuit is grounded, the control signal lamp shall be checked for compliance with 3.14.4.

4.9.5 Headsets and lamp, control signal. - With dc power ON, connect a short circuit across the retransmit control (output) and transmit control circuits at RADIO SET NO. 1 connector (J1). Place HEADSETS 1 switch to TRANSMIT and rotate the function switch of COMM CONT NO. 1 to each of the five numbered switch positions. Observe the response of the CONTROL SIGNAL lamp at each position 1 through 5. Connect a jumper from the primary power return circuit at the RADIO SET NO. 1 connector J1-E to the mike transmit/intercom return circuit at the COMM CONT NO. 2 connector J3-k. Also, connect a jumper from transmit control at the RADIO SET NO. 2 connector J2-h to the mike transmit control circuit at the COMM CONT NO. 2 connector J3-B. Place COMM CONT switch to 2 and HEADSETS 2 switch to TRANSMIT and observe the response of the CONTROL SIGNAL lamp. The requirements of 3.14.5 shall be met.

4.9.6 Control, C-6533( )/ARC (Rec). - Apply a 1000 Hz signal to the intercom audio circuit (interphone line) at the COMM CONT No. 2, connector J3. Connect an eight ohm load across the HEADSET 1 Jack 15. Adjust the output signal of the signal generator to obtain 1.25 volts across the 8 ohm load. Measure the output voltage of the signal generator. Repeat the test at 300 Hz and 6000 Hz. The requirement of 3.14.6 shall be met.

4.9.7 Control, C-6533( )/ARC (Trans). - Connect a 5K to 5 ohm matching network between the microphone high and low inputs at the HEADSETS 1 jack and the signal generator. Adjust the signal generator for an output of 1000 Hz at 0.60 volts ac and place the HOT MIKE switch on. Measure the output of the COMM CONT No.1 at the COMM CONT No.2 connector J3-d and J3-Z (Common). Repeat the test at 300 Hz and 6000 Hz. The requirement of 3.14.7 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.9.8 Radio antenna, (S1) XCVR, atten. - Apply modulated signals of 30, 50, 400 MHz to J12 of the equipment and place Radio Antenna Function switch to XCVR position, measure the output at J11 as each of these frequencies are applied to determine compliance with 3.14.8.

4.9.9 Switch (S1) radio antenna function. - Apply 30 to 80 MHz modulated to J12 and set Radio Antenna Function switch to specified position and terminate J11 with a 50 ohm load. Terminate J14 with a 50 ohm load and measure the output of J13. Move termination of J14 to J13 and measure the output of J14. These measurements shall be made at each frequency specified to determine compliance with 3.14.9.

4.9.10 ADF Antenna, atten. - Set test set switches S2 through S5 to OFF. Apply each frequency 100 kHz and 3.0 MHz to J23, terminate jacks and set switch S1 to each required position when each output is measured. These measurements shall be made at each frequency to determine compliance with 3.14.10.

4.9.11 T/R Antenna, atten. - Connect a coaxial jumper between POWER METER OUTPUT jack J9 and POWER METER INPUT jack J10. Apply 30, 50, 100, 200, 250, 300, 350 and 400 MHz with 50 percent modulation at 1000 Hz, to T/R ANTENNA jack J11. Set RADIO TEST switch in position 4, the attenuation at P3 should be less than 1.0 db. Measure the output of J11 at each frequency to determine compliance with 3.14.11.

4.10 Visual and mechanical inspection. - Prior to and after assembly, the equipment shall be examined for the defects in accordance with MIL-STD-252, and the requirements of 3.19 shall be met.

4.11 Interchangeability. - The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirements of 3.17. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers, and insertable assemblies when such dimensions affect mating of parts.

b. Dimensions of cavities, of the console and other units, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers and mountings.

d. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units, and location of the mating parts on the plug-in assembly or major units.

e. Size and form of special threads.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

4.12 Operational inspection. - The MK-994( )/AR shall be inspected and tested for assurance that all accessories are in place and that the test set controls perform their required functions for compliance with 3.16.

4.13 Quality conformance inspection of preparation for delivery. - Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the packaging requirements of Section 5.

4.14 Rough handling test (preparation for delivery). - When rough handling test in accordance with Specification MIL-P-116 is required by the contract, the operational inspection test 4.12 shall be conducted to determine freedom from operational malfunction caused by rough handling.

5. Preparation for delivery. -

5.1 Preservation and packaging. - Preservation and packaging shall be A or C as specified (See 6.2).

5.1.1 Level A. -

5.1.1.1 Cleaning. - Cleaning shall be accomplished in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. - Drying shall be accomplished in accordance with the applicable procedures of MIL-P-116.

5.1.1.3 Preservation application. - None required.

5.1.1.4 Unit packaging. - Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 and as specified herein.

5.1.1.4.1 Technical literature. - Each Technical literature shall be packaged per Method IC-1.

5.1.1.4.2 Test Facilities Kit, MK-994( )/AR. - Each test kit shall be packaged Method III as follows. Stow and secure all accessories and spare parts within the lid of the test kit case. Close the interior lid and secure fastenings. Cut a sheet of Type I, Class A, cellulosic cushioning material conforming to PPP-C-843. Place the cut sheet over the control and instrument panel of the test kit. Close the case lid and secure fastenings. Cushion the case on all surfaces with cells or pads or both fabricated of fiberboard conforming to PPP-F-320, Type CF, Class weather-resistant variety SW, Grade W5c, designed to protect all projections and absorb the shock of impact encountered in handling and transit. Place the cushioned test facilities kit within a close-fitting fiberboard box conforming to PPP-B-636, W5c. Place the Technical Literature, packaged as specified in 5.1.1.4.1, on top of the contents, directly under the lid of the box. Close the box in accordance with the appendix of the box specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

5.1.2 Level C. - Test Facilities Kit MK-994( )/AR shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2 Packing. - Packing shall be level A, B, or C as specified. Shipping containers for all levels shall be capable of stacking and supporting superimposed loads, during shipment and storage without damaging the container(s) or its contents. (See 6.2)

5.2.1 Level A. -

5.2.1.1 Palletized load. - A quantity of Test Facilities Kit MK-994( )/AR, packaged as specified in 5.1, shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s or V3c. The load shall be "bonded" to the pallet by strapping.

5.2.1.2 Less than palletized load. - When quantities per destination are less than a pallet load, the item packaged as specified in 5.1, shall be waterproofed, with tape conforming to PPP-T-76, in accordance with the taping requirements of the appendix of the box specification. A quantity of the waterproofed containers shall be placed within a close-fitting box conforming to PPP-B-601, overseas type; PPP-B-621, style 4, class 2; or PPP-B-585, style 2 or 3, class 3. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, 3 x 4 inch skids, laid flat, shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment. Closure and strapping shall be in accordance with the applicable container specification or appendix thereto except that metal strapping shall conform to QQ-S-781, type 1, class B.

5.2.2 Level B. -

5.2.2.1 Palletized load. - A quantity of Test Facilities Kit, MK-994( )/AR, packaged as specified in 5.1, shall be palletized as specified in 5.2.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

5.2.2.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Test Facilities Kit MK-994( )/AR, packaged as specified in 5.1, shall be placed within a close-fitting fiberboard box conforming to PPP-B-640, class style E, or PPP-B-636, type CF, class weather-resistant, variety DW. The gross weight of boxes conforming to PPP-B-640 shall not exceed 250 pounds. When the gross weight exceeds 200 pounds, or the container length and width is 48 x 24 inches or more and the weight exceeds 100 pounds, containers will be pallet-mounted on pallets conforming to MIL-P-15011 or MIL-P-3938. Closure shall be in accordance with the appendix of the applicable box specification. Reinforcing shall be by nonmetallic strapping or pressure-sensitive filament tape banding conforming to PPP-S-760, type II and PPP-T-97, type IV respectively; selection of the material and application shall be in accordance with the appendix of the applicable box specification.

5.2.3 Level C. - Test Facilities Kit MK-994( )/AR shall be preserved and packaged in a manner that will afford adequate protection against physical and environmental damage during shipment, handling and limited intransit storage.

5.2.3.1 Palletized load. - A quantity of Test Facilities Kit, MK-994( )/AR packaged as apecified in 5.1, shall be palletized as specified in 5.2.1.1 except that the fiberboard caps shall be class domestic.

5.2.3.2 Less than palletized load. - When quantities per destination are less than a pallet load, a quantity of Test Facilities Kit MK-994( )/AR, packaged as specified in 5.1, shall be packed as specified in 5.2.2.2 except that the fiberboard boxes shall conform to PPP-B-640 and PPP-B-636, class 1 and class domestic, respectively, and reinforcing shall not be required for boxes conforming to PPP-B-636.

5.3 Marking. - In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. - Test Facilities Kit, MK-994( )/AR, in conjunction with standard test equipment is used as an aid in maintenance, providing facilities for bench testing Radio Sets AN/ARC-114( ), AN/ARC-115( ), AN/ARC-116( ), Control, Communication System C-6533( )/AR and Direction Finder Set AN/ARN-89( ) at Direct or General Support Levels and Electronic Shops, Shelter Mounted, Avionics.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

b. Level of preservations, packaging and level of packing required for shipment. (Level A, level B or level C, See Section 5).

c. The specific paragraphs of Section 5 which are applicable to the particular procurement.

d. Preproduction inspection. -

(1) Three (3) sample units of each item cited in Section 1 are generally required so that lengthy environmental tests can be completed on one sample unit while complete performance measurements can be made on the second sample unit. (See 3.2)

(2) Preproduction pack(s) as follows:

(a) Makeup of pack(s).

(b) Number of each kind of pack to be submitted.

(c) Inspection to be performed thereon. (Including rough handling and functional tests).

e. Marking and shipping of samples.

f. Place of final inspection.

g. Technical literature required. (See 3.18)

h. Quantity of tools and running spare parts required.

i. Submission of the statement of treatment referenced in 3.12 should be as specified in the solicitation/contract. This statement should be submitted to the contracting officer.

6.3 Nomenclature. - The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example; MK-994W/AR. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.4 Group C inspection. - Approval to ship may be withheld, at the discretion of the Government, pending results of the corrective action specified in the contract.

6.5 Definition of Level B. - Level B packaging requirements detailed herein are designed to furnish protection against corrosion, deterioration, and physical damage during domestic shipments; covered storage for a period of three years or less; and distribution within the continental limits of the United States.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.

6.6 Location of Operational Inspection. - It is desirable that the operational inspection 4.12 be performed at a location that will minimize handling (which might cause damage to the units) after this inspection. It is recommended that the entire lot (including the sample units) be inspected immediately prior to packaging.

6.7 Verification Inspection. - Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive Quality Assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

Custodian:

Preparing Activity
Army EL
Project No. 6625-A481

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:18Z
Check the source to verify that this is the current version before use.